Paired-Reference, Key-Free Indication of a Public SynthID-Text Instance

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In plain English. A watermark is meant to be checked with a secret key. This report asks whether, given matched marked and unmarked samples but no keys, an auditor can still tell that Google DeepMind's public SynthID-Text demo is on. At the level of prompt groups, yes. On one 128-token file at a pre-set threshold, no. That is key-free under a strong paired oracle, not a polynomial-time distinguisher, and not a detector validated for an unfamiliar web document. Figure 3 is the two-grain picture; Figure 5 and Table 13 are the preregistered $H_{\mathrm{w}} = 12$ attenuation, not a second headline. The public `get_gvals` docstring does not match the code (Section 1). The counts are in the abstract below.

Abstract

With matched labeled marked and unmarked reference generations, and with detector keys withheld at scoring time, empirical next-token count tables rank held-out prompt-group means on Google DeepMind's public SynthID-Text instance (`public-deepmind-30`). Exploratory in-domain ranking is **36/36** families (hits reader; file AUC 0.930). Under a predeclared leave-one-family-out protocol on 100 new GPT-2 families, lock A Witten–Bell interpolate ranking is **99/100** (mean paired difference 0.520; one family misses). The corrected original 12-family hard-reader diagnostic is **9/12** group wins.

These group successes are not standalone file detection. On the original 96 files the prespecified zero threshold yields 25 true positives, 23 false negatives, 22 true negatives, and 26 false positives: sensitivity **25/48**, specificity **22/48**, and **47/96** correct decisions (48.96% balanced accuracy). Threshold-free file AUC is 0.590. On 20 occupancy-unseen marked files the official keyed mean still exceeds the laboratory threshold 0.55 (**20/20** positive-control sensitivity; unmarked files in that slice are not a matched negative class). The method is key-free, not reference-free: it requires a strong paired oracle on one short-context public instance. It does not recover keys, has not been validated or calibrated for unfamiliar web documents, and does not refute Christ et al. (2024) or Zhang et al. (2024).

Keywords: statistical text watermarking, SynthID-Text, paired-reference auditing, key-free indication, empirical likelihood ratio.

1 Introduction

Statistical watermarks for language models bias next-token sampling with a secret keyed function, then test whether a finished string contains more of that structure than chance (Kirchenbauer et al., 2023; Aaronson and Kirchner, 2023; Dathathri et al., 2024). SynthID-Text is a tournament member of that family, deployed at scale and released as a public reference implementation (Dathathri et al., 2024; Google DeepMind, 2024b). Official detection needs the instance keys. Those keys exist in the public reference code; this report withholds them from the auditor by design.

The public `get_gvals` docstring says g-values come from "the lowest three bits" of the n-gram keys. The installed code does 12 linear-congruential mixes and then (hash $\gg 30$) mod 2.

This report trusts the code. That mismatch is a documentation bug in the public DeepMind checkout, not a laboratory rewrite.

Isolated-headline scope. The locked isolated headline is the original-12 hard last-4 matrix: sensitivity **25/48**, specificity **22/48**, accuracy **47/96**, file AUC 0.590. Later isolated counts—other generators, mixins, windows, nested thresholds, or occupancy-free readouts—do not replace that headline. That is not **25/48** as a refrain: the rule is this paragraph and table footnotes.

If a third party cannot or will not use the vendor's keys, verification is an open practical question (Google DeepMind, 2024a; Gloaguen et al., 2025; Wang et al., 2026).

This report asks a scoped version of that question, on one public instance, under a strong oracle:

> *Given matched marked and unmarked generations from the same prompts, and no detector keys at scoring time, can an auditor tell whether `public-deepmind-30` is on—as a ranking of prompt groups, and as a decision on one 128-token file?*

The two grains must not be collapsed. On the measured corpora, paired-reference group indication can be strong even when a prespecified isolated-file threshold is uninformative. A slogan of the form "key-free detection works" or "key-free detection fails" is false for this measurement.

1.1 Contribution

Independent work already asked whether a third party can detect a sampling watermark without the vendor's keys (Section 2). Gloaguen et al. (2025) test a generator property with black-box queries. Wang et al. (2026) verify finished strings from paired watermarked and unwatermarked references, the closest published analog. The increment here is a two-grain measurement under a strong paired oracle on DeepMind's public mixin: keys are withheld at scoring, but matched twins are given. Group ranking can succeed while a prespecified isolated threshold fails. Count tables on finished strings are ordinary polynomial-time statistics; lock A **99/100** is not a complexity distinguisher against Christ et al. (2024) or Zhang et al. (2024). Cross-construction measurements are contextual comparisons, not a second headline.

In order:

1. A paired-reference empirical log-likelihood ratio. Decision-time inputs are finished strings. Keys, `hash_iv`, g-values, and the official score are unused.

2. A two-grain measurement on `public-deepmind-30`. Exploratory in-domain ranking **36/36**; prospectively specified GPT-2 lock A **99/100**; corrected original hard-reader diagnostic **9/12**. Isolated zero-threshold decisions on the original set are **47/96** correct (sensitivity **25/48**).

3. Cross-generator native refitting on the same 100 prompts: DistilGPT-2 lock B **88/100** (one tie), Qwen2-1.5B lock B **95/100**. Rankpath drops more than opening poshits on both generators (paired McNemar on GPT-2-win losses).

4. Artifact controls: a truncated-context overcount correction, a Laplace occupancy versus observed-token split, and a key-free argmax perturbation that moves the official mean close to 0.5 at a large rewrite cost.

5. A preregistered longer-history replication: public keys at $H_{\mathrm{w}} = 12$ (`ngram_len` = 13). Official matching scores remain above 0.55 on all 400 marked 100-family files. Interpolate last-4 group ranking falls from lock A **99/100** to **76/100**. Exact next-token occupancy is lower than under public $H_{\mathrm{w}} = 4$ (Table 13). Isolated 489/800 is a different corpus from the original-12 **47/96**.

What this is not: key recovery; a web-crawler detector; a calibrated isolated-file classifier; a production Gemini or Anthropic-key measurement; a complexity-theoretic distinguisher against Christ et al. (2024); a mixing attack against Zhang et al. (2024); a new watermark construction.

The intended venue is a technical report. The work is a bounded empirical note on one public short-context instance, not a flagship detection result.

2 Related Work

The audit question is not new. Table 1 places this report against the closest families. The precise novelty dimension is simple empirical count-table auditing under paired labeled references for a fixed public instance, without training a proxy model.

Sampling watermarks. Kirchenbauer et al. (2023) bias sampling with a keyed green list. Kuditipudi et al. (2024) give a distortion-free inverse-transform construction. SynthID-Text reweights the language model's top-K with keyed g-values (Dathathri et al., 2024). This repository's `score` path calls the public `detector_mean`; it does not reimplement it. The public `get_gvals` docstring does not match the installed code (Section 1). Omidi et al. (2026) analyse keyed scores. That is not the present indicator.

Undetectability and strong watermarking. Christ et al. (2024) prove cryptographic undetectability for a stated PRF construction. Zhang et al. (2024) prove impossibility of *strong* watermarking given a quality oracle and a mixing oracle. This laboratory did not instantiate that scheme and did not run those oracles. Lock A **99/100** is not a distinguisher against their objects. Isolated **47/96** is a different grain, not a restoration of those theorems.

Stealing and public verifiability. Jovanović et al. (2024) learn spoof/remove models from API samples. Concurrent stealing includes Wu and Chandrasekaran (2024) and Pang et al. (2024). We do not steal keys. Duan et al. (2025) wrap keyed detectors in a zero-knowledge proof; the detector still uses the key. DeepMind's public-verification request remains open (Google DeepMind, 2024a).

Third-party detection. Gloaguen et al. (2025) detect a watermarking scheme as a generator property. SRI Lab applied those tests to a local SynthID deployment (SRI Lab, ETH Zurich, 2024). That is not scoring a finished string against count tables. Anthropic's announced Claude mark is a future external test (Anthropic, 2026); production keys are not public, and `score` on `public-deepmind-30` cannot read that instance.

3 Preliminaries and Threat Model

3.1 Notation

Table 2 is used throughout. Implementation field `lr` equals Λ. Token slices $[a{:}b)$ are half-open generated-token index intervals. Unless stated, history preceding a slice remains visible only in the absolute-history protocol (Section 4.4).

3.2 SynthID-Text on the public instance

An autoregressive model supplies $P_0(x_t \mid x_{<t})$. SynthID-Text does not insert hidden characters. It biases which token is sampled (Dathathri et al., 2024).

Table 1: Task-level comparison. TTP-Detect costs are from Wang et al. (2026) (their §5.1 and Table 2), not remeasured here. Count-table size is from the committed 100-family interpolate dump. There is no shared-benchmark TPR. Simpler is not automatically a better detector.

Axis	TTP-Detect (Wang et al., 2026)	This report
Oracle	Paired watermarked / unwatermarked references	Same: matched twins
Keys at test	Unused	Unused (public keys exist and are withheld)
Fitted object	LoRA Qwen2.5-3B proxy + Qwen3-1.7B; relative tests	Sparse next-token counts and Λ (0 neural parameters)
Training references	6000 C4 paired completions (their paper)	$N \times 4$ twins (100 families here)
Training hardware	2×A100 80GB (their paper)	CPU; no accelerator
Detection-time queries	$N = 16$ marked + $N = 16$ unmarked refs ($\approx$ 6.14s)	None (finished-string lookup)
TTP-side scoring	$<$ 2s GPU modules (their Table 2)	CPU mean Λ over observed 4-grams
Persisted reader	3B proxy + 1.7B scorer	33.2 MB JSON; 108454 / 116491 contexts
Decision unit	Finished-string tests in that paper	Prompt-group mean and one file
Demonstrated scope	Their corpora and scheme mix	One public SynthID instance

Table 2: Reserved symbols. Older laboratory notes used k for top-k, count-table history, and mask length; that collision is retired here.

Symbol	Meaning
$\mathcal{V}$	Generator tokenizer vocabulary
V_{obs}	Lidstone / Witten–Bell support (cardinality of the observed-token union)
K	Language-model top-K (40 in this laboratory)
H_{w}	Watermark hash history: $\texttt{ngram\_len} - 1 = 4$
h	Count-table history length (default 4)
τ	Isolated-file decision threshold (headline $\tau = 0$)
m	Number of leading scored positions masked in an ablation
M	Draws per prompt family (4)
T	Generated tokens per file (128)
$\Lambda(x)$	Mean empirical log-likelihood ratio of file x
D_p	Prompt-level paired difference of group means

The public reference configuration is `public-deepmind-30`: `ngram_len` = 5, hence watermark hash history $H_{\mathrm{w}} = 4$; 30 public keys; `context_history_size` = 1024. The implementation hashes the recent context with a linear congruential generator. As already noted in Section 1, g-values are (hash $\gg$ 30) mod 2 after 12 LCG mixes, not the docstring's "lowest three bits." g-values are binary.

Only the language model's top-K candidates are reweighted ($K = 40$; temperature 1.0). Repeated context hashes are masked so they do not add repeated evidence. The first H_{w} generated tokens cannot form a complete *official* scored 5-gram.

The official mean detector averages unmasked g-values of observed tokens. Under unmarked generation from this fixed instance, $\mathbb{E}[g] = 1/2$ over hash outputs along the string. Distortion-freeness in expectation over *random keys* is a claim of the scheme (Dathathri et al., 2024), not a theorem proved here. A deployed instance fixes one key set.

Table 3 is the keyed positive control on the original 12 prompts (first draw). The `Prompt` column is `detector_mean` of the prompt string itself. The prompt is not stamped; those scores sit near 1/2. Every marked continuation sits well above 0.55; unmarked continuations stay near

1/2. The cutoff 0.55 is a laboratory convenience for counting keyed positives. It is not a published DeepMind decision rule.

3.3 The paired auditor

Definition 3.1 (Key-free paired auditor)**.** The auditor $\mathcal{A}$ has black-box access to two configurations of the same generator and tokenizer:

(i) unmarked G_0, producing $x^{(u)} \sim P_0(\cdot \mid p)$;

(ii) marked G_w, producing $x^{(m)} \sim P_w(\cdot \mid p)$ under a fixed public instance.

It holds a paired training corpus of finished strings

$$\mathcal{D}_{\text{train}} = \{(p_i, x_{i,j}^{(m)}, x_{i,j}^{(u)})\}$$

across N prompt families and M draws per family. It does *not* use the instance keys, `hash_iv`, g-value tables, or the official scoring oracle at decision time. The keys are public in the reference implementation and are withheld by design. Official `score` is a positive control that the mixin is on, not the contribution.

This is a strong oracle: a laboratory that can toggle the mixin, or a before/after collection of matched outputs. The method is **key-free**. It is **not reference-free** and **not fully blind**. In this repository, `blind` means leave-one-prompt-out on those twins.

Marked and unmarked draws are *prompt-matched*, not seed-paired. For family offset r and draw j, marked and unmarked files use distinct `torch.manual_seed` values (Section 4.1). Do not treat a marked/unmarked pair as a shared sampling stream.

3.4 Two evaluation grains

Task 1: prompt-group ranking. For a held-out prompt p with M marked and M unmarked draws, declare a win if

$$D_p = \frac{1}{M}\sum_{j=1}^{M} \Lambda(x_{p,j}^{(m)}) - \frac{1}{M}\sum_{j=1}^{M} \Lambda(x_{p,j}^{(u)}) > 0. \tag{1}$$

Ties ($D_p = 0$) are not awarded. Both means may be negative: a win then means the unmarked group is more negative, not that any marked file is positive.

Task 2: isolated-file decision. For one file x of $T = 128$ generated tokens,

$$\hat{y}(x) = \mathbf{1}\{\Lambda(x) > \tau\}, \qquad \tau = 0 \text{ unless named as post hoc.} \tag{2}$$

No twin is shown at test time. The headline isolated result uses the prespecified $\tau = 0$, not a Youden threshold (Youden, 1950) fit on the same stem. The primary isolated statistic is the 2×2 confusion matrix and the balanced accuracy $\frac{1}{2}(\mathrm{TP}/(\mathrm{TP}+\mathrm{FN}) + \mathrm{TN}/(\mathrm{TN}+\mathrm{FP}))$. On the original 96 files the two classes each have 48 files, so that quantity equals ordinary accuracy $(\mathrm{TP} + \mathrm{TN})/96 = \mathbf{47/96}$. The binomial interval $[0.386, 0.594]$ is for 47 of 96 correct decisions under a shared Bernoulli model, not a general interval for balanced accuracy with separate sensitivity and specificity. Sensitivity and specificity keep their own intervals. Prompt-clustered sign-flip of balanced accuracy is reported below.

The independent ranking unit is the prompt family. Leave-one-prompt-out tables share training mass: each LOFO fit on the 100-family lock uses the other 99 families, so consecutive models share 98 of 99 training families. Results are dependent through that overlapping fit-set.

Table 3: Official keyed `detector_mean` on the original 12 GPT-2 prompt pairs (first draw). Positive control; uses keys.

Stem	Prompt string	Unmarked gen.	Marked gen.
harbour	0.500	0.508	0.621
night-bus	0.498	0.496	0.633
library	0.493	0.499	0.633
market	0.479	0.505	0.630
kitchen	0.509	0.506	0.614
station	0.520	0.498	0.627
rain	0.515	0.491	0.617
letter	0.513	0.507	0.622
workshop	0.498	0.491	0.605
office	0.518	0.506	0.647
garden	0.484	0.502	0.629
ferry-queue	0.524	0.499	0.616

Clopper–Pearson intervals (Clopper and Pearson, 1934) on win counts are descriptive under an independent Bernoulli model that the design only approximates. File-level label-shuffle permutation p-values (Ernst, 2004) ignore pairing and clustering; they are descriptive. Exact McNemar tests on discordant prompt-family signs (McNemar, 1947) are clustered descriptions of already collected scores, not a second preregistration.

Declared operating criteria. The primary group endpoint is the prompt-family win count of (1). The primary isolated endpoint is balanced accuracy at the prespecified $\tau = 0$, reported with the full 2×2 matrix. Sensitivity, specificity, and precision are companions of that matrix, not substitutes for it. File AUC (Fawcett, 2006) is a secondary, threshold-free ranking statistic; it does not test whether $\tau = 0$ is calibrated. For abstaining readers, coverage and conditional performance on files with $n > 0$ are reported separately: $n = 0$ is not counted as a confident negative.

The 12, 36, and 100 prompt sets are designed collections, not random draws from a stated population. Intervals describe a hypothetical repeated-sampling process on those fixed prompts and the laboratory generator, not inference to English at large.

4 Method

4.1 Paired generation

All twins use Hugging Face generators with the public mixin. Length is matched at 128 new tokens (`min_new_tokens=max_new_tokens`). Temperature is 1.0, top-$K = 40$, and the pad token equals EOS. Prompts are plain strings; Qwen2-Instruct is not wrapped in a chat template. EOS does not truncate early: every file has 128 new tokens.

Seed allocation for family offset r, $M = 4$, base seed s: marked draw $j \in \{0, 1, 2, 3\}$ uses $s + (2M + 2)r + 2j$; unmarked draw j uses one more. Marked and unmarked never share a seed.

Default Hub identifiers: `gpt2`, `distilgpt2`, `Qwen/Qwen2-1.5B-Instruct`. Hub revisions were not recorded when the twins were generated. That does not affect the published scores: key-free tables and official `detector_mean` read the committed strings, not a Hub snapshot. It would matter for bitwise re-generation of new twins. The laboratory machine's current `gpt2` cache is `607a30d783dfa663caf39e06633721c8d4cfcd7e`. DistilGPT-2 and Qwen snapshots were not retained. Future `pair` runs should pass a Hub `revision` (`pair --hub-revision`). `transformers==4.57.6`. SynthID-Text is the public reference package installed with `pip` `install -e . --no-deps` (the public `synthid-text` checkout). Exact frozen dumps and commands are in Appendix A.

Corpora used in the headline tables:

- Original 12 scene prompts (one everyday scene each), GPT-2, 4 draws. Headline 12-LOO hard reader, after the truncated-context recount.
- 36 in-domain GPT-2 topics $\times$ 4 draws. Exploratory hits reader.
- 100 new English one-line scene prompts, disjoint as strings from the 36, seed `20260901`. Prospective lock A/B/C.
- The same 100 prompts on DistilGPT-2 and Qwen2-1.5B-Instruct (Phase B). Native tables per generator. Lock A interpolate was not run on those twins.

4.2 Empirical mean log-likelihood ratio

Fit two next-token count tables on the training twins. Generated token 0 is skipped by default (no nonempty last-h context without the prompt). Prompt tokens are not available as context unless `prompt-context` is named. Smoothing is Lidstone with $\alpha = 0.5$ (Lidstone, 1920). For a context c and token y,

$$\hat{P}(y \mid c) = \frac{C(c, y) + \alpha}{\sum_{y'} C(c, y') + \alpha V_{\text{obs}}}, \tag{3}$$

where V_{obs} is the size of the union of next tokens observed in the marked and unmarked training unigrams, not $|\mathcal{V}|$. Logs are natural. If an evaluation token $y \notin V_{\text{obs}}$, the implementation does not map y to `UNK` and does not skip the position. The unigram term is $(C(\varepsilon, y) + \alpha)/(n + \alpha V_{\text{obs}})$ with $C(\varepsilon, y) = 0$, so occupancy mass $\alpha/(n + \alpha V_{\text{obs}})$ is assigned outside the training-support partition; the evaluation conditional is therefore not a normalized distribution over $V_{\text{obs}} \cup \{y\}$. Interpolate tries Witten–Bell backoff first; the occupancy formula still applies at the empty context. If a context is seen in one table and unseen in the other, the unseen side uses unigram Lidstone (hard) or Witten–Bell backoff (interpolate).

The file score is the *mean* per scored position:

$$\Lambda(x) = \frac{1}{n} \sum_{t \in \mathcal{I}(x)} \log \frac{\hat{P}_{\text{m}}(x_t \mid c_t)}{\hat{P}_{\text{u}}(x_t \mid c_t)}, \tag{4}$$

where $\mathcal{I}(x)$ are scored generated indices and $n = |\mathcal{I}(x)|$. If $n = 0$, $\Lambda(x) = 0$. For the headline hard reader that assignment is a numerical convention, not an abstention state: after skipping token 0, every 128-token file has $n > 0$. Hats are essential: the auditor never observes P_w. Equation (4) has the Neyman–Pearson likelihood-ratio form (Neyman and Pearson, 1933), with estimated rather than known conditionals; it is not claimed to be an optimal test of P_w versus P_0.

At generated index i, the scored context is the up-to-h previous *generated* tokens. Window [0:4) therefore contains indices $\{0, 1, 2, 3\}$; with token 0 skipped, three events remain, with context lengths $1, 2, 3$ rather than a full last-4. That is why [0:4) is an opening measurement rather than four complete $h = 4$ events.

Fitting stores each real suffix length $1, \ldots, \min(h, i)$ once. An earlier bug stored a truncated opening h times (Section 7.1).

4.3 Readers

hard. Equation (3) at the longest available context, with unigram fallback if that key is empty. Primary original-12 diagnostic.

hits. Score a position only if the exact last-h context was observed in *both* training piles; else skip it. Exploratory 36-topic ranking reader.

poshits. Namespace counts by position bucket $i//B$. Lock B uses `fit-prefix` $= 4$ and $B = 1$, so each of the first four generated tokens has its own table.

postokhits. As poshits, but a position contributes only if the observed next token occurred under that namespaced context (occupancy-free). Positions may abstain. If every position abstains, $n = 0$ and $\Lambda = 0$; coverage and conditional performance are reported separately and zeros are not treated as confident negatives.

interpolate (lock A). Witten–Bell interpolation (Witten and Bell, 1991). Let n_c be the count mass of context c and r_c the number of distinct next tokens. If c is unseen or $n_c = 0$, recurse to the shortened suffix. Otherwise

$$\hat{P}(y \mid c) = \frac{n_c}{n_c + r_c} \cdot \frac{C(c, y)}{n_c} + \frac{r_c}{n_c + r_c} \cdot \hat{P}(y \mid c^-),$$

with c^- the context with the oldest token dropped, and with Lidstone unigram at the empty context. Probabilities are clipped below at 10^{-18} before the log. Unseen next tokens are therefore scored by backoff: interpolate nested recall is not occupancy-free recall.

rankpath (lock C). Replace token identity with a five-symbol unmarked-LM rank alphabet on the unmarked top-$K = 40$: symbol 0 if the chosen token misses top-K; 1 if rank 1 (argmax); 2 if ranks 2–3; 3 if ranks 4–10; 4 if ranks 11–40. Those five bins partition $\{0\} \cup \{1, \ldots, 40\}$. Isolated rankpath skips generated token 0 (no unmarked-LM context without the prompt), so `fit-prefix` $= 4$ yields three rank symbols. Rankpath uses observed-token backoff on the symbol sequence (`include_first` on symbols). It needs the unmarked language model at scoring time and still does not use keys.

Hash pooling is a laboratory splitmix of token ids. It is not SynthID's secret hash and is not used in the headline tables.

Occupancy. If a context is seen in training but the evaluation next token is not, add-α still assigns a nonzero log ratio from the two occupancy denominators. That mass is not an observed-token hit. `postokhits` removes it.

4.4 Windows

Slice $[a{:}b)$ denotes generated indices i with $a \leq i < b$. Two implementations exist.

- *Reindexed.* The slice is copied out as a new string. Token 0 of that substring is skipped, and contexts do not include tokens before a. The first committed H2 dump used this behaviour.
- *Absolute.* The full string is retained. Only indices in $[a{:}b)$ are averaged; preceding tokens remain as context. Later probes refuse to reindex a window as a new sequence.

H2 reports both. The reindexed dump is not overwritten.

Algorithm 1 (leave-one-family-out ranking and isolated scores).
Input: families $p = 1, \ldots, N$, M marked and M unmarked files each; reader R.
for $p = 1$ to N **do**
Fit $\hat{P}_{\mathrm{m}}, \hat{P}_{\mathrm{u}}$ with R on all families except p.
Score $\Lambda(x_{p,j}^{(m)})$ and $\Lambda(x_{p,j}^{(u)})$ for $j = 1, \ldots, M$.
Ranking win if $D_p > 0$ (Equation (1)).
Isolated label $\hat{y}(x) = \mathbf{1}\{\Lambda(x) > 0\}$ (hard headline).
end for

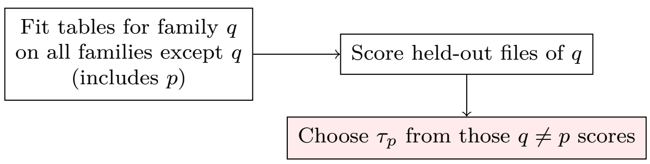
Figure 1: Tables for family p never contain p's own files. They do contain every other family. Consecutive LOFO models share $N-2$ of $N-1$ training families, so the N wins are dependent.

4.5 Leave-one-family-out

For each family p, tables are fit on all other families and the $2M$ held-out files of p are scored. That is the evaluation topology of **9/12**, **36/36**, **99/100**, Distil **88/100**, and Qwen **95/100**. Pseudocode is Figure 1.

This is a frozen *algorithm and evaluation protocol*, not a frozen fitted detector. Lock A **99/100** is leave-one-family-out on the 100 new families after the reader definition was locked. It is not application of tables fit on the original 12.

Nested Youden is post hoc. A later readout chooses, for family p, a Youden threshold on the other families' already-held-out scores (Youden, 1950). Those other scores were produced by models that trained on p: when family $q \neq p$ was scored, its leave-one-out tables included p. The threshold for p therefore sees mass from p. The procedure is a threshold nest, not second-level nested cross-validation, and is not an unbiased held-out classifier (Figure 2). Isolated headlines use $\tau = 0$. Nested Youden counts stay out of detector-performance comparisons.



Unbiased nested evaluation would remove p before *any* fitting or threshold selection. This laboratory did not run that procedure.

Figure 2: Nested Youden leakage. Outer-test family p can influence the scores used to pick τ_p. Headline isolated decisions do not use this threshold.

4.6 Confirmatory locks

Before generating the 100 new GPT-2 families, three readers were named (protocol SHA `7001489`; prompts committed in `294dba5`):

- **Lock A.** Interpolate last-4 (`--methods interpolate`, `--context-len 4`).
- **Lock B.** Opening poshits (`--methods poshits --fit-prefix 4 --pos-bucket 1`).
- **Lock C.** Opening rankpath (`--rankpath --fit-prefix 4 --pos-bucket 1`).

Hypothesis 4.1 (H1)**.** On new GPT-2 prompt families, lock A prompt ranking (1) is the primary confirmatory endpoint. H1–H3 are directed predictions with descriptive evaluation; no α or rejection rule was predeclared.

Hypothesis 4.2 (H2)**.** Under lock A, window [0:4) ranks more families than window [16:32) (opening sufficient in-domain).

Hypothesis 4.3 (H3)**.** Native-refit rankpath ranking is less stable across generator families than native-refit poshits: on DistilGPT-2 and Qwen2-1.5B, the number of GPT-2 ranking wins lost under native refitting is larger for rankpath than for poshits. Both readers are refit natively; this is not frozen GPT-2 transfer.

H1–H3 are predictions about prompt ranking. Isolated $\tau = 0$ is logged and is not H1. Phase B does not run lock A. Phase B tables are *native*: Distil tables are fit on Distil twins, Qwen tables on Qwen twins. H3 therefore measures variation in within-generator learnability, not frozen GPT-2 transfer.

Table 4 records the topology of every headline.
No headline hyperparameter was selected on its evaluation corpus. Reader flags were frozen in protocol SHA `7001489` before the 100-family generation. Isolated headlines use $\tau = 0$, not a threshold fit on the scored files.

4.7 A named non-inferiority margin (demoted)

A laboratory robustness switch declares a ranking win when $\bar{\Lambda}_{\mathrm{m}} + 0.02 > \bar{\Lambda}_{\mathrm{u}}$, i.e. $D_p > -0.02$. On the original 12 this moves ferry-queue and yields 10/12. That rule is not the Bernoulli trial of Equation (1). Under a symmetric no-effect distribution, $P(D_p > -0.02)$ exceeds 1/2, so a Clopper–Pearson interval around 10/12 that excludes 1/2 does not establish above-chance performance under the margin. The 0.02 tolerance was not a pre-registered primary endpoint of lock A. It is reported only as a descriptive sensitivity analysis and is absent from the abstract.

5 Why a Fixed Key Can Leave a Trace

The following is a mechanism sketch, not a tight theorem. Empirical Λ uses $\hat{P}_{\mathrm{m}}$ and $\hat{P}_{\mathrm{u}}$, not P_w.

Let G be a random key and g^* the deployed instance. Distortion-freeness in expectation over keys does not imply equality for one reused key:

$$\mathbb{E}_G[P_G(\cdot \mid c)] = P_0(\cdot \mid c) \quad \not\Rightarrow \quad P_{g^*}(\cdot \mid c) = P_0(\cdot \mid c). \tag{5}$$

Tournament sampling on a fixed g-table reweights top-K in a context-dependent way. Repeated draws from the same prompt reuse short contexts ($H_{\mathrm{w}} = 4$ on this mixin). Matched samples can therefore produce $\hat{P}_{\mathrm{m}} \neq \hat{P}_{\mathrm{u}}$ on occupied contexts without revealing the key.

Conditionally on the fitted tables, the within-family contrast satisfies

$$\mathrm{Var}\Big(D_p \mid \hat{P}_{\mathrm{m}}, \hat{P}_{\mathrm{u}}\Big) = O(1/M)$$

if the $2M$ draws are independent with finite per-file variance. That statement is not unconditional: all held-out scores share estimated tables. We do not claim an exponential Hoeffding bound, and we do not claim that isolated $T = 128$ classification is impossible in general.

Scaling hypotheses, not production facts. As H_{w} grows, exact held-out context overlap should fall and sample demand should rise. As K grows, observations per context-token outcome spread more thinly. Lower-order Witten–Bell backoff may retain a coarser signal that is no longer a reused 4-gram footprint. Lengthening this mixin to $H_{\mathrm{w}} = 12$ attenuates group ranking and lowers exact next-token occupancy (Section 9, Table 13). A Kirchenbauer green-list mixin on the original 12 prompt strings ranks interpolate last-4 **12/12** with mostly Witten–Bell backoff (Section 9). That is a shorter hash window than $H_{\mathrm{w}} = 4$, not a longer-history result. An

Table 4: Train / test topology. "Prospective" means the reader and flags were named before the evaluation corpus was generated. Clopper–Pearson intervals in later tables are descriptive.

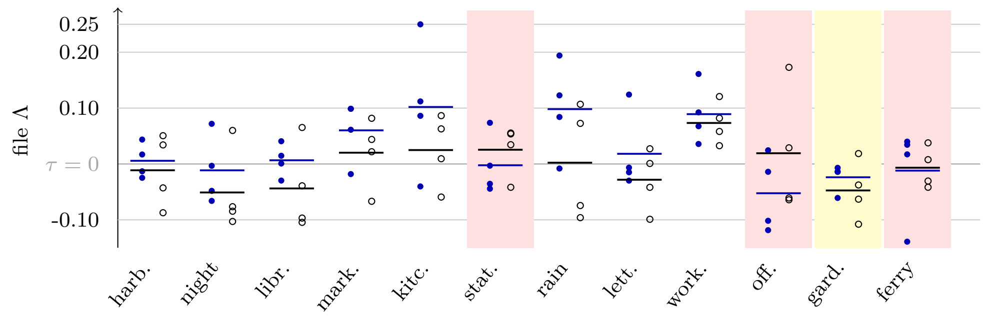
Result	Status	Tables fit on	Threshold	Unit
12 hard 9/12	Exploratory; locked after recount	Other 11 of the original 12	ranking >	prompt
12 hard isolated	Same tables	Same	$\tau = 0$ (prespecified)	file
36 hits 36/36	Exploratory LOO	Other 35 of the 36	ranking >	prompt
GPT-2 lock A 99/100	Prospective protocol; LOO refit	Other 99 of the new 100	ranking >	prompt
Distil/Qwen B, C	Prospective flags; native LOO	Other 99 of that generator's 100	ranking >	prompt
Keyed 12/12, 100/100, 70/100	Positive control (keys)	none (no count table)	first-draw marked mean > unmarked mean of official detector	prompt, first draw

Aaronson mixin (freeze SHA `747f3cd`) ranks interpolate last-4 **11/12** on the original 12 strings with isolated balanced accuracy 56/96.

Christ et al. and Zhang et al. apply to different constructions and oracles (Section 2). Isolated chance accuracy at $\tau = 0$ is compatible with a weak single-file test. Population ranking of matched twins is outside their single-challenge games as stated. Neither sentence is a theorem about tournament sampling in their sense.

6 Results

Figure 3 is the paper's central picture: four marked and four unmarked file scores per original stem, group means, and the zero threshold. Garden wins the group comparison with no marked file above zero. Station, office, and ferry-queue lose the group comparison and still contribute five isolated true positives.



• marked file ○ unmarked file short bars: group means.

Yellow (garden): ranking win, 0/4 marked files above τ. Rose (station, office, ferry-queue): ranking loss with isolated TPs.

Figure 3: Original 12 families, 12-LOO hard last-4. Group ranking uses the two bars; isolated decisions use the zero line. Garden: negative–negative ranking win, 0/4 marked files above τ. Station, office, and ferry-queue lose ranking and still contribute five isolated true positives.

6.1 Headline two-grain comparison

Table 5 is the main pane. Official keyed detection is the positive control, with a different decision unit from the key-free locks (Table 4). Group ranking is the key-free claim. Isolated $\tau = 0$ is the full confusion matrix, not sensitivity alone.

Table 5: Two-grain results. Status marks exploratory versus prospective locks versus keyed controls. Descriptive Clopper–Pearson 95% intervals invert the binomial (Clopper and Pearson, 1934) and ignore shared tables. Isolated TP/TN is marked $\Lambda > 0$ / unmarked $\Lambda \le 0$ at $\tau = 0$. Precision is $\mathrm{TP}/(\mathrm{TP} + \mathrm{FP})$. Balanced accuracy is $\frac{1}{2}(\mathrm{sens} + \mathrm{spec})$; on this 48+48 set it equals accuracy 47/96. The interval $[0.386, 0.594]$ is binomial for 47 of 96 correct decisions, not a general BA interval. File AUC permutation is a 2000-shuffle of file labels (Ernst, 2004); it ignores prompt clustering and pairing and is not inferential support for $\tau = 0$.

Protocol	Status	Keys	Unit	Prompt-group ranking	Isolated $\tau = 0$
Official mean, original 12 (first draw)	control	yes	prompt	12/12	—
Prompt-group ranking					
12-LOO hard last-4	diagnostic	no	prompt	**9/12** $[0.428, 0.945]$	TP/TN 25/22
Hits, 36×4 LOO	exploratory	no	prompt	**36/36** $[0.903, 1.000]$; AUC 0.930	136/72
GPT-2 100×4 lock A LOO	prospective	no	prompt	**99/100** $[0.946, 1.000]$; AUC 0.898	352/290
DistilGPT-2 lock B native LOO	prospective	no	prompt	**88/100** $[0.800, 0.936]$ (1 tie)	—
Qwen2-1.5B lock B native LOO	prospective	no	prompt	**95/100** $[0.887, 0.984]$	—
Isolated original 12, hard last-4, $\tau = 0$					
Confusion matrix	diagnostic	no	file	—	TP 25, FN 23, TN 22, FP 26
Sensitivity / specificity	diagnostic	no	file	—	25/48 $[0.372, 0.667]$, 22/48 $[0.314, 0.608]$
Precision	diagnostic	no	file	—	25/51 $[0.348, 0.634]$
Accuracy / BA (primary isolated)	diagnostic	no	file	—	**47/96** (48.96%); BA equals accuracy here
Prompt-clustered BA permutation	diagnostic	no	file	—	stem-flip $p = 0.703$ (compatible with chance)
File AUC	diagnostic	no	file	—	0.590 (file-shuffle $p = 0.040$, unclustered)

Hard last-4 **9/12** includes 1/2 in its descriptive interval. Isolated accuracy / balanced accuracy **47/96** also includes 1/2. Mean marked Λ on those 48 files is $+0.023$; mean unmarked is -0.002. The distributions overlap. Precision at $\tau = 0$ is 25/51 ($[0.348, 0.634]$). A prompt-clustered sign-flip of marked/unmarked group labels within each stem (20,000 draws, conservative $(n_{\ge} + 1)/(N + 1)$) yields $p = 0.703$ for balanced accuracy at least as large as 47/96; a within-stem 4/4 reshuffle yields $p = 0.672$. Zero-threshold accuracy is compatible with chance under those clustered nulls. A weakly informative score can still have AUC above 0.5: AUC asks whether marked files tend to rank above unmarked files across thresholds (Fawcett, 2006); the file-shuffle AUC permutation $p = 0.040$ ignores prompt clustering and pairing and does not test whether threshold zero is calibrated. The defensible isolated statement is: *the prespecified zero threshold does not yield useful isolated-file decisions on this set; threshold-free file ranking is weak (AUC* 0.590*) and requires independent confirmation.*

6.2 Ranking and isolated sign disagree on stems

Table 6 lists every original-12 family. Night-bus is a second negative–negative ranking win. The population grain can succeed without any isolated file testing positive.

6.3 In-domain 36×4

Hits on 36 topics × 4 draws, leave-one-prompt-out, ranks **36/36** (file AUC 0.930; descriptive file-level permutation $p \approx 0.0005$). Isolated $\tau = 0$ on that reader is 136/144 marked and only 72/144 unmarked at or below zero: many unmarked files are also positive, and ranking succeeds because marked means are larger (mean difference 1.27). That isolated operating point is not the original-12 hard headline.

A 16-token prefix already ranks **34/36** (AUC 0.916). Window [0:4) matches that ranking (**34/36**, AUC 0.917). Window [16:32) is near chance (**22/36**, AUC 0.549). On this corpus an opening is sufficient for in-domain ranking. Count-table history $h = 5$ (one token longer than watermark history $H_\mathrm{w} = 4$) does not beat $h = 4$ (hits 35/36, AUC 0.912). Those two lengths are different objects: `ngram_len` = 5 is the mixin's 5-gram; $h = 5$ is a longer next-token table.

6.4 Confirmatory 100 GPT-2 families

Protocol freeze before generation; analysis flags unchanged after the first run. Official first-draw keyed comparison (marked `detector_mean` > unmarked, one file each) is **100/100** (descriptive interval $[0.964, 1.000]$). Minimum marked mean ≈ 0.576; maximum unmarked mean ≈ 0.519.

Lock A leave-one-family-out interpolate ranking is **99/100**. Figure 4 shows the 100 paired differences D_p. Mean D_p is 0.520 (median 0.504). Every win has $D_p \geq 0.133$; three wins lie in $[0.13, 0.20)$. The miss, stem 088, has $D_p = -0.238$. Reporting only 99 signs would hide that most wins are large. One win has both group means negative.

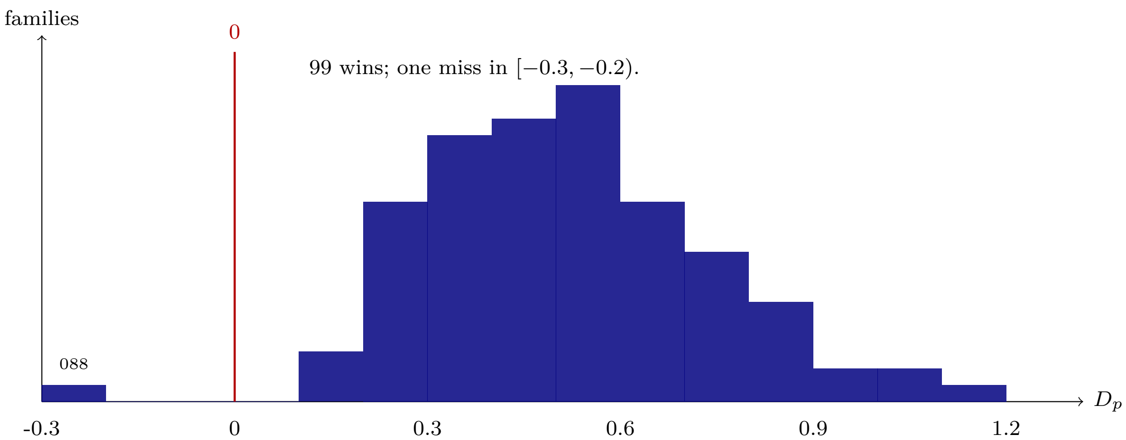


Figure 4: Lock A prompt-level paired differences D_p on 100 GPT-2 families (leave-one-family-out interpolate). Bins of width 0.1. The mass sits well to the right of zero.

H1 holds. Nested-by-stem Youden on lock A is 322/400 vs. 338/400. Lock B nested Youden is 392/400 vs. 382/400, with occupancy (198/400 unmarked files have $n_\mathrm{used} = 0$). Those file counts are not an isolated-file detector on the original 12.

6.5 Cross-generator Phase B

Same 100 prompts, native DistilGPT-2 and Qwen2-1.5B-Instruct tables. Locks B and C only. Table 8 separates the keyed first-draw control from key-free four-draw ranking.

Distil keyed means: marked ≈ 0.559, unmarked ≈ 0.496; minimum marked ≈ 0.433. First-draw Distil **70/100** is marked mean > unmarked mean (six exact ties, not awarded). On this

Table 6: 12-LOO hard last-4 stem means. Isolated TP: marked files with $\Lambda > 0$. Isolated TN: unmarked files with $\Lambda \leq 0$.

Stem	Marked mean	Unmarked mean	Win?	TP	TN
harbour	+0.0057	−0.0113	yes	2/4	2/4
night-bus	−0.0114	−0.0510	yes	1/4	3/4
library	+0.0066	−0.0438	yes	3/4	3/4
market	+0.0602	+0.0202	yes	3/4	1/4
kitchen	+0.1020	+0.0249	yes	3/4	1/4
station	−0.0023	+0.0255	no	1/4	1/4
rain	+0.0982	+0.0023	yes	3/4	2/4
letter	+0.0182	−0.0282	yes	1/4	2/4
workshop	+0.0892	+0.0734	yes	4/4	0/4
office	−0.0524	+0.0192	no	1/4	2/4
garden	−0.0238	−0.0474	yes	0/4	3/4
ferry-queue	−0.0119	−0.0067	no	3/4	2/4
Total			**9/12**	**25/48**	**22/48**

Table 7: Confirmatory 100×4 GPT-2, leave-one-family-out. Isolated $\tau = 0$ counts are in-family and use a different operating point from the original-12 hard matrix. Nested Youden is post hoc (Section 4.5).

Lock	Reader	Prompt wins	File AUC	Isolated TP/TN at $\tau = 0$
A	interpolate last-4	**99/100**	0.898	352/290
B	opening poshits	**100/100**	0.980	393/344
C	opening rankpath	**96/100**	0.822	314/302

sample the same 70 families are the ones with marked mean > 0.55; that coincidence is not a reason to equate the two rules. Qwen keyed means: marked ≈ 0.580, unmarked ≈ 0.500. Five Qwen marked files sit at or below 0.55 and still beat their unmarked twins, so the two keyed rules diverge there. Ground truth for key-free probes remains mixin on/off, not the keyed pass rate.

H3 as a paired drop contrast, using prompts as the unit. A drop is a GPT-2 ranking win that is not a win under native refitting (ties count as non-wins). Distil lock B loses 12 GPT-2 wins and gains none (net 100 → 88). Distil lock C loses 31 GPT-2 wins and gains 3 (net 96 → 68). Of those Distil losses, 6 prompts drop only on B, 25 only on C, and 6 on both. Exact one-sided McNemar $P(X \geq 25 \mid \mathrm{Bin}(31, 1/2)) \approx 0.00044$. Qwen lock B loses 5 and gains none (net 100 → 95); lock C loses 15 and gains 3 (net 96 → 84). Qwen B-only drops 2, C-only 12, both 3; $P(X \geq 12 \mid \mathrm{Bin}(14, 1/2)) \approx 0.0065$. Native-refit rankpath ranking lost more GPT-2 wins than native-refit poshits on both generators. This is a post hoc clustered description of already collected scores, not a second preregistration. Lock B starts at 100/100 and lock C at 96/100 on GPT-2, so raw percentage-point drops also differ by ceiling.

Count tables are not a tokenizer-universal isolated detector. GPT-2 learned scorers transferred to Distil/Qwen are chance. Native Distil opening rankpath on the original 12 is compatible with chance (**8/12**, AUC 0.579) even though Distil official first-draw on those 12 is **12/12**.

6.6 Leftover versus covered isolated true positives

The leftover split is the cleanest evidence that group indication and standalone file reading are different capabilities. Mixed occupancy-free tables from the 100 one-liners and 36 longer-scene families cover 28 of the original 48 files; 20 files remain zeros (leftover). Crossing that set with the headline 25 isolated true positives gives Table 9.

Table 8: Two different endpoints. Keyed column: first-draw official `detector_mean`, marked mean > unmarked mean (one file each). Lock B/C: four-draw empirical-Λ ranking. Distil keyed 70/100 is a paired first-draw keyed-control win rate (marked mean > unmarked mean), not a standalone official detector pass rate. Historical JSON printed Distil 89/100 and 69/100 under a non-strict comparison; locked counts treat a tie as not a win.

Generator (100 families)	Keyed first-draw	Lock B poshits	Lock C rankpath
GPT-2	100/100	100/100	96/100
DistilGPT-2	70/100	**88/100** (1 tie)	68/100 (1 tie)
Qwen2-1.5B-Instruct	100/100	**95/100**	84/100

Table 9: Headline isolated $\tau = 0$ split by mixed occupancy-free opening coverage. Leftover last-4 is compatible with chance. Official keyed leftover **20/20** is positive-control sensitivity on marked files at the laboratory threshold 0.55, not a two-class detection claim (no unmarked leftover column). Ranking-loss TPs (station, office, ferry-queue) are all leftover.

Slice	n	TP ($\Lambda > 0$)	TN ($\Lambda \leq 0$)	Keyed marked > 0.55
Occupancy-covered	28	15/28	11/28	28/28
Leftover (unseen openings)	20	10/20	11/20	**20/20**
Original 12×4	48	25/48	22/48	48/48

Official keyed mean on leftover-20 marked files is 0.624. That is positive-control sensitivity at 0.55, not evidence that the keyed detector separates leftover files the empirical reader cannot decide: unmarked leftover files are not scored as a matched negative class here.

Further leftover unions, Distil occupancy-free transfer, and mask-m tails are catalogued in Appendix B. They refine the same boundary; they do not change the headline two-grain claim. Table 13 is in-family leave-one-out interpolate occupancy on new $H_{\mathrm{w}} = 12$ twins, not a leftover union on the original 12.

7 Ablations and Mechanistic Perturbation

7.1 Truncated-context overcount

Early logs stored every order $1, \ldots, h$ even when fewer tokens existed, so a one-token opening was counted h times. After storing each real suffix once, hard last-4 ranking moved from 10/12 to **9/12** and isolated sensitivity from 29/48 to **25/48**. Prompt ranking survived. Pre-fix JSON is historical.

7.2 Laplace occupancy versus observed tokens

On a 4-token new-topic gate, poshits isolated $\tau = 0$ is 39/48 vs. 41/48. Occupancy-free `postokhits` is 16/48, with decided precision 1.000 when $n_{\text{used}} > 0$. Isolated observed-token recall on named opening readers equals train opening-atom overlap. That identity does not create a denser isolated-file detector. Comparing marked recall alone with 25/48 is not a sufficient evaluation rule. A low-recall, high-specificity abstaining reader is a different operating point.

7.3 H2 windows

H2 asked whether lock A window [0:4) outranks [16:32) on the 100 GPT-2 families. Reindexed: [0:4) **99/100** (AUC 0.885), [16:32) **89/100** (AUC 0.689). Absolute-history: [0:4) **99/100** (AUC 0.885), [16:32) **87/100** (AUC 0.695). Opening still outranks mid-file in-family. Paired McNemar on the absolute windows is 86/13/1/0 (both / opening only / mid only / neither); $P(X \geq 13 \mid \mathrm{Bin}(14, 1/2)) \approx 0.00092$. That is not a second preregistration.

Out-of-family, the same interpolate tables are not front-loaded on the longer-scene files (reindexed [0:4) is 7/12; tails 9/12).

7.4 Argmax perturbation

Tournament sampling only reweights the unmarked top-K. Snapping each generated token to the unmarked argmax, using original prefixes, is a key-free *mechanistic perturbation*. It needs the unmarked language model, not the keys. About 60–90 of 128 tokens flip. On all 48 marked original-12 files, official mean moves 0.6216 → 0.4994. The procedure is not a fluent rewrite, not self-consistent regeneration, and not a practical removal attack.

8 Limitations

The public instance is a short-context ($H_{\mathrm{w}} = 4$) demo, not a proprietary production key. Group ranking needs matched draws per prompt. Isolated classification of one 128-token file at $\tau = 0$ is compatible with chance on the original set. File AUC 0.590 is weak and was not a confirmatory endpoint.

The auditor is strong: marked and unmarked twins, finished strings, a toggleable mixin. Closest published finished-string analog is Wang et al. (2026); this repository does not reimplement TTP-Detect on a shared corpus. Table 1 quotes their published proxy and A100 costs against this laboratory's committed count-table dump. That is a resource contrast, not a shared-benchmark TPR.

Nested Youden is not fully nested. The 0.02 margin is not a valid chance test. Prompt sets are designed, not sampled from a stated population. Lock A **99/100** is prospectively specified leave-one-family-out, not evaluation of one frozen fitted table on a later corpus. Distil keyed 70/100 is a paired first-draw keyed-control win rate, not the four-draw key-free endpoint. Isolated balanced accuracy at $\tau = 0$ is not distinguishable from chance under a prompt-clustered sign-flip.

How count-table coverage scales with longer watermark history is measured on this public mixin: $H_{\mathrm{w}} = 12$ attenuates group ranking from lock A **99/100** to **76/100** on the same 100 prompt strings (different twins). Interpolate occupancy is lower under $H_{\mathrm{w}} = 12$ than under public $H_{\mathrm{w}} = 4$ on both prompt frames (Table 13). Production Gemini and Anthropic keys were not measured. Hub revisions were not recorded at generation; published scores read committed strings (Section 4.1). Multi-key replication of the public mixin is not started. Isolated-file research is not finished.

9 Preregistered extensions

The preregistered longer-history replication is freeze SHA `b70986d` (`research/PROTOCOL-next-longctx.md`), named in the laboratory logbook before generation (Table 12). The same commit also names the absolute-history mask remasure (`research/PROTOCOL-isolated-mask-absolute.md`); they share a freeze SHA, not a dump. It lengthens the public mixin hash history to $H_{\mathrm{w}} = 12$ without editing the `synthid-text` checkout.

A Kirchenbauer green-list mixin is freeze SHA `8371406` (`research/PROTOCOL-next-kgw.md`; Hugging Face `WatermarkLogitsProcessor` defaults, seed 20260904, `mixin=kgw`), named before generation. Scoring those twins with SynthID `detector_mean` would be a harness bug.

DistilGPT2 with the same Kirchenbauer defaults is frozen in `research/PROTOCOL-next-kgw-distil.md` (freeze SHA `1540d3c`). Distil interpolate last-4 is **12/12** (isolated **85/96**); hard is **11/12**. Qwen2-1.5B Kirchenbauer on the same 12 strings is frozen in `research/PROTOCOL-next-kgw-qwen.md` (freeze SHA `45a75c9`). Qwen interpolate last-4 is **12/12** (isolated **68/96**); hard is **8/12**.

DistilGPT2 Kirchenbauer on the 100 one-liners is frozen in `research/PROTOCOL-next-kgw-distil.md` (freeze SHA `4fad227`). Distil interpolate last-4 is **100/100** (isolated **683/800**); hard is

82/100. Qwen2-1.5B Kirchenbauer on the 100 one-liners is freeze SHA `ed9fb20` (`research/PROTOCOL-next-kgw-qwen-100.md`; `experiments/2026-09-04-pair-qwen-100x4-kgw`). Observed counts are in Table 10 and the paragraph below. Isolated-file detection remains unfinished.

Table 10: Qwen2-1.5B Kirchenbauer 100-family measurement (protocol SHA `ed9fb20`). Flags were locked before generation. Observed key-free counts are dump-backed. That lock is not **25/48**.

Item	Value
Mixin	Hugging Face `WatermarkLogitsProcessor` (`mixin=kgw`)
Generator	`Qwen/Qwen2-1.5B-Instruct`
Hub revision	`ba1cf1846d7df0a0591d6c00649f57e798519da8`
`greenlist_ratio` / `bias` / `hashing_key`	0.25 / 2.0 / 15485863
`seeding_scheme` / `context_width`	`lefthash` / 1
Prompt frame	100 one-liners, $T = 128$, seed 20260904, four draws
Official control	matching `WatermarkDetector` $z > 3$
Reader	hard and interpolate last-4; `used_keys=false`
Fold topology	leave-one-family-out
Primary isolated endpoint	balanced accuracy at $\tau = 0$, with the 2×2 matrix

Official first-draw $z > 3$ is **90/100** (unmarked 0/100). Interpolate last-4 is **96/100** (isolated **620/800**); hard is **63/100**. Occupancy is **4858** seen versus **96740** unseen. That lock is not **25/48**. Whether interpolate **96/100** is a body reader or an opening residual is freeze SHA `e270546` (`research/PROTOCOL-next-kgw-qwen-100-windows.md`). Interpolate [0:4) is **84/100**; [64:128) is **97/100** (AUC **0.785**). That is not compatible with chance, and not $H_{\mathrm{w}} = 12$ **50/100**. JSON: `experiments/2026-09-04-probe-qwen-100x4-kgw-windows`.

Isolated rankpath geography does not replace the locked hard matrix. Opening rankpath on the original 12 is **11/12**, isolated **41/48 vs 35/48**; the same reader on [4:16) is **7/12**, isolated **20/48 vs 22/48** (AUC 0.430). H-rplm-d fails as a raw count on Distil-LM openings (**32/48 vs 31/48**). Equality of any later isolated count with **25/48** is not a win. Isolated-file detection remains unfinished.

Table 11: Unmarked-LM rankpath cemetery on GPT-2-family twins (12-LOO). Isolated counts are marked $\Lambda > 0$ versus unmarked $\Lambda \leq 0$ at $\tau = 0$. None of these isolated counts replace the original-12 hard headline **25/48**. Protocol files live under `research/`.

Reader	Freeze	Isolated $\tau = 0$	Ranking
Distil-LM on GPT-2, opening	`d8e6f7f`	32/48 vs 31/48	10/12
medium-LM on GPT-2, opening	`d8e6f7f`	31/48 vs 32/48	10/12
medium native, opening	`2577771`	22/48 vs 30/48	6/12
GPT-2-on-medium, opening	`336a1fd`	20/48 vs 32/48	8/12
Distil-on-medium, opening	`b3fd331`	30/48 vs 31/48	11/12
GPT-2-on-Distil, opening	`d62c732`	24/48 vs 27/48	6/12
medium-on-Distil, opening	`571d4f1`	30/48 vs 33/48	9/12
GPT-2 body [4:16)	`dbc61c5`	20/48 vs 22/48	7/12
Distil-LM body [4:16)	`68b0514`	24/48 vs 23/48	6/12
medium-LM body [4:16)	`3ea80e4`	27/48 vs 28/48	9/12
Distil native body [4:16)	`468a66b`	25/48 vs 30/48	9/12
medium native body [4:16)	`37a2c43`	20/48 vs 30/48	6/12
GPT-2-on-Distil body [4:16)	`08b89ee`	26/48 vs 21/48	4/12
Distil-on-medium body [4:16)	`1b4c541`	25/48 vs 33/48	11/12
GPT-2-on-medium body [4:16)	`e677a6c`	28/48 vs 30/48	8/12
medium-on-Distil body [4:16)	`a550cb6`	32/48 vs 25/48	9/12
GPT-2 mid [16:32)	`14afbd5`	23/48 vs 26/48	7/12

Protocol notes: `PROTOCOL-isolated-rankpath-lm.md`, `PROTOCOL-isolated-rankpath-m12.md`, `PROTOCOL-isolated-rankpath-g2m.md`, `PROTOCOL-isolated-rankpath-d2m.md`, `PROTOCOL-isolated-rankpa` `PROTOCOL-isolated-rankpath-m2d.md`, `PROTOCOL-isolated-rankpath-body.md`, `PROTOCOL-isolated-rankp` `PROTOCOL-isolated-rankpath-mbody.md`, `PROTOCOL-isolated-rankpath-d12body.md`, `PROTOCOL-isolated-`

PROTOCOL-isolated-rankpath-g2dbody.md, PROTOCOL-isolated-rankpath-d2mbody.md, PROTOCOL-isolate
PROTOCOL-isolated-rankpath-m2dbody.md, PROTOCOL-isolated-rankpath-mid.md. Window dumps live under experiments/2026-09-04-probe-*-w4-16/. The frozen slice is always window-4-16/, not the unwindowed fit-prefix-16 file score (isolated-headline scope).

An Aaronson–Kirchner exponential-minimum mixin (freeze SHA 747f3cd, seed 20260905) ranks interpolate last-4 **11/12** (isolated **56/96**); hard is **12/12**. On the 100 one-liners interpolate last-4 is **100/100** (isolated **608/800**). The protocol note is research/PROTOCOL-next-aaronson.md. DistilGPT2 $H_{\mathrm{w}} = 12$ on the same 12 strings is frozen in research/PROTOCOL-next-longctx-distil.md (freeze SHA bae6d81; experiments/2026-09-04-pair-distil-12x4-ngram13). Distil $H_{\mathrm{w}} = 12$ interpolate last-4 is **9/12** (isolated **49/96**); hard is **6/12**. Official first-draw mean$>$ 0.55 is 12/12. Unmarked first-draw is 0/12 above 0.55. Occupancy is 175 seen versus 11994 unseen (opening [0:4) is 111 versus 177). DistilGPT2 $H_{\mathrm{w}} = 12$ on the 100 one-liners is frozen in research/PROTOCOL-next-longctx-distil-100.md (freeze SHA d891622; experiments/2026-09-04-pair-distil-100x4-ngram13). Distil $H_{\mathrm{w}} = 12$ interpolate last-4 is **88/100** (isolated **557/800**); hard is **89/100**. Official first-draw mean$>$ 0.55 is 98/100. Unmarked first-draw is 0/100 above 0.55. Occupancy is 11182 seen versus 85493 unseen (opening [0:4) is 2036 versus 364). Qwen2-1.5B $H_{\mathrm{w}} = 12$ on the same 12 strings is frozen in research/PROTOCOL-next-longctx-qwen.md (freeze SHA d7303a2; experiments/2026-09-04-pair-qwen-12x4-ngram13). Qwen $H_{\mathrm{w}} = 12$ interpolate last-4 is **4/12** (isolated **41/96**); hard is **4/12**. Official first-draw mean$>$ 0.55 is 11/12 (library 0.515). Unmarked first-draw is 0/12 above 0.55. Occupancy is 65 seen versus 12127 unseen (opening [0:4) is 41 versus 247). DistilGPT2 Aaronson on the same 12 strings is frozen in research/PROTOCOL-next-aaronson-distil.md (freeze SHA 9bdf12a; experiments/2026-09-04-pair-distil-12x4-aaronson). Distil Aaronson interpolate last-4 is **7/12** (isolated **0/48** marked true positives, BA **48/96**); hard is **7/12** (isolated **56/96**). Occupancy is 196 seen versus 11996 unseen (opening [0:4) is 133 versus 155). Unmarked first-draw is 1/12 above 3.0. DistilGPT2 Aaronson on the 100 one-liners is frozen in research/PROTOCOL-next-aaronson-distil-100.md (freeze SHA bf05759; experiments/2026-09-04-pair-distil-1
Distil Aaronson interpolate last-4 is **96/100** (isolated **601/800**); hard is **91/100**. Official first-draw $z > 3$ is 71/100. Unmarked first-draw is 3/100 above 3.0. Occupancy is 28824 seen versus 61305 unseen (opening [0:4) is 2048 versus 352). Qwen2-1.5B Aaronson on the same 12 strings is frozen in research/PROTOCOL-next-aaronson-qwen.md (freeze SHA 1171d5c; experiments/2026-09-04-pair-qwen-12x4-aaronson). Qwen Aaronson interpolate last-4 is **12/12** (isolated **12/48** marked true positives, BA **60/96**); hard is **12/12** (isolated **72/96**). Unmarked first-draw is 0/12 above 3.0. Occupancy is 457 seen versus 11735 unseen (opening [0:4) is 127 versus 161). Qwen2-1.5B Aaronson on the 100 one-liners is frozen in research/PROTOCOL-next-aaronson-qwen-100.md (freeze SHA a761a7d; experiments/2026-09-04-pair-qwen-100x4
Qwen Aaronson interpolate last-4 is **100/100** (isolated **616/800**); hard is **97/100**. Official first-draw $z > 3$ is 99/100. Unmarked first-draw is 0/100 above 3.0. Occupancy is 8750 seen versus 92842 unseen (opening [0:4) is 1470 versus 930). Qwen2-1.5B $H_{\mathrm{w}} = 12$ on the 100 one-liners is frozen in research/PROTOCOL-next-longctx-qwen-100.md (freeze SHA 636765c; experiments/2026-09-04-pair-qwen-100x4-ngram13). Qwen $H_{\mathrm{w}} = 12$ interpolate last-4 is **76/100** (isolated **474/800**); hard is **74/100**. Official first-draw mean$>$ 0.55 is 91/100. Unmarked first-draw is 0/100 above 0.55. Occupancy is 3535 seen versus 98064 unseen (opening [0:4) is 1092 versus 1308). Whether $H_{\mathrm{w}} = 12$ interpolate **76/100** is a weaker body reader or an opening residual is named in research/PROTOCOL-next-longctx-windows.md (freeze SHA 8283d1f). Opened: interpolate [64:128) **50/100** (AUC **0.501**) versus public $H_{\mathrm{w}} = 4$ **93/100** (AUC **0.726**); opening [0:4) **86/100**. JSON: experiments/2026-09-04-probe-100x4-ngram13-windows, experiments/2026-09-04-probe-100x4-public-w64-128.

Original 12, $H_{\mathrm{w}} = 12$**.** The original-12 replication uses new twins from those prompt strings (seed 20260903). Official means with matching ngram_len = 13 are above 0.55 on all 48 marked

Table 12: Preregistered $H_{\mathrm{w}} = 12$ measurement, locked before generation (protocol SHA `b70986d`). Observed counts follow in the paragraphs below.

Locked before generation	Value
Mixin / H_{w}	public-deepmind-30 keys, `ngram_len` = 13 ($H_{\mathrm{w}} = 12$)
Generator / tokenizer	`gpt2`
Prompt frame	original 12 scene strings, then 100 one-liners
Reference size	12×4 then 100×4 matched twins, $T = 128$, seed 20260903
Reader	hard and interpolate last-4 (lock A analog)
Fold topology	Algorithm 1 leave-one-family-out
Zero / abstention	$n = 0 \Rightarrow \Lambda = 0$; coverage reported
Primary group metric	D_p sign count, strict >
Primary isolated endpoint	balanced accuracy at $\tau = 0$, with the 2×2 matrix

files (first draw already 12/12). Unmarked files stay near 1/2 (0/48 above 0.55). Leave-one-family-out interpolate last-4 ranks **6/12** (AUC 0.541; $\tau = 0$ is 20 TP, 28 FN, 31 TN, 17 FP; balanced accuracy 51/96). Hard last-4 is also **6/12** (AUC 0.544; 22 TP, 26 FN, 30 TN, 18 FP; balanced accuracy 52/96). Both prompt-win Clopper–Pearson intervals include 1/2 (interpolate clustered permutation $p = 0.247$). That is below the public-mixin hard diagnostic **9/12** on these prompt strings (different twins, $H_{\mathrm{w}} = 4$). Isolated $\tau = 0$ on the new twins is 52/96, a different corpus from the original-12 **47/96**. Leave-one-family-out interpolate occupancy is in Table 13: 160 exact next-token events versus 269 on the public $H_{\mathrm{w}} = 4$ twins. Isolated interpolate remains 20/48. Hard prompt wins are market, station, rain, letter, workshop, and ferry-queue. Interpolate wins library, market, station, letter, workshop, and ferry-queue. Five stems win on both readers. Mean paired file differences are small (0.009 hard, 0.021 interpolate).

One hundred families, $H_{\mathrm{w}} = 12$. The longer-history 100-family readout uses the same lock on 100 one-liners (named in the logbook before generation; not Distil/Qwen Phase B). Official matching `ngram_len` = 13 is above 0.55 on **400/400** marked files (first draw already **100/100**). Leave-one-family-out interpolate last-4 ranks **76/100** (AUC 0.666; $\tau = 0$ is 267 TP, 133 FN, 222 TN, 178 FP; balanced accuracy 489/800), below public-mixin lock A **99/100** on these prompt strings. Hard is **66/100** (AUC 0.579; balanced accuracy 436/800). The original-12 $H_{\mathrm{w}} = 12$ set **6/12** is imprecisely estimated at $n = 12$; the 100-family readout is a cross-run contrast, not a one-factor proof that longer history drops ranking. Prompt-clustered sign permutation on interpolate D_p is $p \approx 0.0005$ (2000 prompt-mean sign shuffles). Isolated 489/800 is a different corpus from the original-12 **47/96**. Mean D_p is 0.156 (median 0.164; range $[-0.386, 0.602]$). Occupancy on this frame is 5878 versus 10158 exact events (Table 13). Isolated interpolate remains 267/400. Figure 5 sits much closer to zero than lock A (Figure 4, mean 0.520).

Table 13: Leave-one-family-out interpolate occupancy: exact next-token events versus Witten–Bell backoff. File LRs match the interpolate holdouts. Occupancy is a mechanistic intermediate, not an isolated-file detector. Opening rows sum to 96×3 and 800×3 (token 0 skipped). Full-file seen+unseen is a few events short of $N \times 127$ (6 on original-12 $H_{\mathrm{w}} = 12$; 11 on public original-12; 98 and 89 on the 100-family frame): the atoms decoder counts interpolate traces and omits a position when context selection returns none. We report observed traces, not a padded 127-slot grid.

Slice	$H_{\mathrm{w}} = 12$ seen	public $H_{\mathrm{w}} = 4$ seen	$H_{\mathrm{w}} = 12$ unseen	public unseen
Original 12, full file	160	269	12026	11912
Original 12, [0:4)	71	84	217	204
100 families, full file	5878	10158	95624	91353
100 families, [0:4)	1287	1633	1113	767

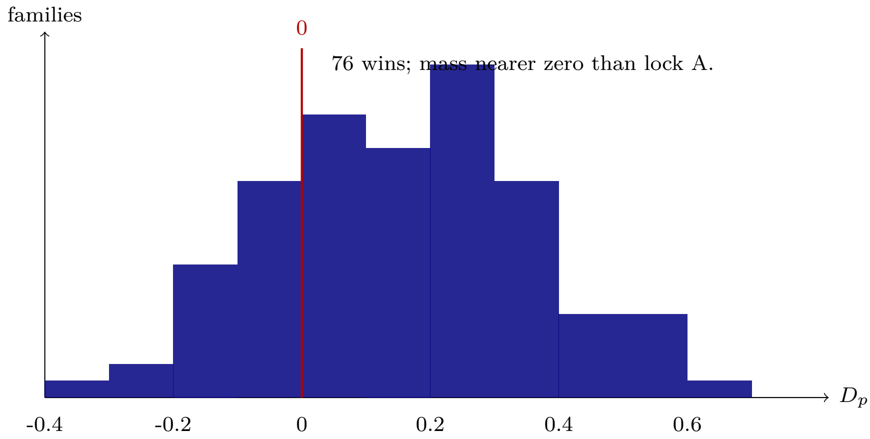


Figure 5: $H_\mathrm{w} = 12$ interpolate prompt-level paired differences D_p on the same 100 prompt strings (new twins; leave-one-family-out). Bins of width 0.1. Compare Figure 4.

Original 12, Kirchenbauer green-list. The different-mixin arm uses Hugging Face Kirchenbauer defaults (`lefthash`, context_width $= 1$, $\gamma = 0.25$, $\delta = 2.0$; freeze SHA `8371406`; seed 20260904; Hub SHA `607a30d783dfa663caf39e06633721c8d4cfcd7e`). Official matching z-scores are above 3.0 on all 48 marked files (first draw already 12/12). Unmarked first-draw is 0/12 above 3.0. Leave-one-family-out interpolate last-4 ranks **12/12** (AUC 0.947; $\tau = 0$ is 44 TP, 4 FN, 41 TN, 7 FP; balanced accuracy 85/96; mean $D_p = 0.437$). Hard last-4 is also **12/12** (AUC 0.703; 35 TP, 13 FN, 25 TN, 23 FP; balanced accuracy 60/96). Exact last-4 occupancy is 114 seen versus 12071 unseen (opening [0:4) is 40 versus 248), below public $H_\mathrm{w} = 4$ original-12 occupancy 269 seen. Interpolate **12/12** is therefore not an exact 4-gram copy detector; Witten–Bell backoff carries the mass. context_width $= 1$ is shorter than public $H_\mathrm{w} = 4$, not a longer-history result. Isolated **85/96** is a different construction from the original-12 SynthID **47/96**.

One hundred families, Kirchenbauer green-list. Same freeze, same Hub SHA, seed 20260904, named `8f09aa6` before generation. Official matching z-scores are above 3.0 on **100/100** first marked files (min 6.61). Leave-one-family-out interpolate last-4 ranks **100/100** (AUC 0.982; $\tau = 0$ is 376 TP, 24 FN, 371 TN, 29 FP; balanced accuracy 747/800; mean $D_p = 0.745$, every family $D_p > 0$). Hard last-4 is **62/100** (AUC 0.573; balanced accuracy 440/800). Exact last-4 occupancy is 4557 seen versus 96991 unseen (opening [0:4) is 1044 versus 1356), below public $H_\mathrm{w} = 4$ 100-family occupancy 10158 seen and below $H_\mathrm{w} = 12$ 5878 seen. Interpolate **100/100** is therefore not an exact 4-gram copy detector. It is a different construction from lock A **99/100** and from the original-12 SynthID **47/96**.

Original 12, DistilGPT2 Kirchenbauer. Same mixin defaults, generator `distilgpt2`, freeze SHA `1540d3c`, seed 20260904. Official matching z-scores are above 3.0 on all 12 first marked files. Interpolate last-4 ranks **12/12** (AUC 0.947; 42 TP, 6 FN, 43 TN, 5 FP; balanced accuracy 85/96; mean $D_p = 0.504$). Hard is **11/12** (AUC 0.780; 66/96). Occupancy is 130 seen versus 11972 unseen. Hub revision was unpinned at generation; published scores read committed strings.

Original 12, Qwen2-1.5B Kirchenbauer. Same mixin defaults, generator `Qwen/Qwen2-1.5B-Instruct`, freeze SHA `45a75c9`, seed 20260904, Qwen tokenizer on probe and atoms. Official matching z-scores are above 3.0 on all 12 first marked files. Interpolate last-4 ranks **12/12** (AUC 0.814; 35 TP, 13 FN, 33 TN, 15 FP; balanced accuracy 68/96; mean $D_p = 0.224$). Hard is **8/12** (AUC 0.566; 50/96). Occupancy is 84 seen versus 12108 unseen. Hub revision was unpinned at generation. Group ranking holds; isolated balanced accuracy is below Distil and GPT-2 Kirchenbauer **85/96**.

One hundred families, DistilGPT2 Kirchenbauer. Same mixin defaults, generator `distilgpt2`, freeze SHA `4fad227`, seed 20260904. Dump `experiments / 2026-09-03-pair-distil-100x4-kgw`. Hub revision `2290a62682d06624634c1f46a6ad5be0f47f38aa`. Official matching z-scores are above 3.0 on **100/100** first marked files (min 6.81). Unmarked first-draw is 16/100 above 3.0 (newline loops scored fully green). Interpolate last-4 ranks **100/100** (AUC 0.915; 346 TP, 54 FN, 337 TN, 63 FP; balanced accuracy 683/800; mean D_p = 0.587, every family $D_p > 0$). Hard is **82/100** (AUC 0.666; 502/800). Occupancy is 16170 seen versus 71541 unseen (opening [0:4) is 1800 versus 600). The top opening atom is a newline loop. First-draw newline-only files are 40/100 marked and 12/100 unmarked. Group ranking matches GPT-2 Kirchenbauer **100/100** on these strings; isolated balanced accuracy is below GPT-2 Kirchenbauer **747/800**.

Original 12, Aaronson–Kirchner exponential-minimum. Laboratory sampler, freeze SHA `747f3cd`, seed 20260905, Hub SHA `607a30d783dfa663caf39e06633721c8d4cfcd7e`, dump `experiments/2026-09-04-pair-12x4-aaronson`. Official matching z-scores are above 3.0 on all 12 first marked files (min 10.84). Unmarked first-draw is 0/12 above 3.0. Interpolate last-4 ranks **11/12** (AUC 0.955; 8 TP, 40 FN, 48 TN, 0 FP; balanced accuracy 56/96; mean $D_p = 1.738$; loses station). Hard is **12/12** (AUC 0.976; 72/96). Occupancy is 573 seen versus 11618 unseen. Nine of eleven interpolate ranking wins have no isolated true positive.

One hundred families, Aaronson–Kirchner exponential-minimum. Same freeze SHA `747f3cd`, seed 20260905, dump `experiments/2026-09-04-pair-100x4-aaronson`. Official matching z-scores are above 3.0 on **100/100** first marked files (min 7.91). Unmarked first-draw is 0/100 above 3.0. Interpolate last-4 ranks **100/100** (AUC 0.986; 208 TP, 192 FN, 400 TN, 0 FP; balanced accuracy 608/800; mean D_p = 4.368). Hard is **95/100** (AUC 0.950; 744/800). Occupancy is 25167 seen versus 76418 unseen. Forty-eight of one hundred interpolate ranking wins have no isolated true positive.

Interpretation. If group ranking had survived when short-context reuse should drop, the work would begin to support a transferable audit method. The observed drop from lock A **99/100** to **76/100** is consistent with the public instance's reusable 4-gram footprint being a boundary of the group-level result. Leave-one-family-out interpolate occupancy is lower under $H_\mathrm{w} = 12$ than under public $H_\mathrm{w} = 4$ on both prompt frames and in the opening window (Table 13). On the original 12 the opening is mostly backoff; on the 100 one-liners it is still mostly exact copies, fewer than under public $H_\mathrm{w} = 4$. The original-12 **6/12** is small-n. Kirchenbauer `lefthash` on the same prompt strings goes the other way: original-12 interpolate **12/12** and 100-family interpolate **100/100** (isolated 747/800) with *less* exact last-4 occupancy than public $H_\mathrm{w} = 4$ or $H_\mathrm{w} = 12$. That matches a one-token green-list plus Witten–Bell backoff, not a universal detector. Neither count is a theorem about production Gemini.

10 Conclusion

With matched labeled marked/unmarked reference generations and no detector keys at scoring time, empirical count-table auditors rank marked prompt-group means above unmarked means in 36/36 exploratory in-domain families and 99/100 prospectively generated same-register GPT-2 families under a predeclared leave-one-family-out protocol. The corrected original hard-reader diagnostic is 9/12. On the original 96 individual files, the prespecified zero threshold yields 25/48 sensitivity, 22/48 specificity, precision 25/51, and 47/96 correct decisions, while threshold-free file AUC is 0.590. Prompt-clustered permutation of balanced accuracy is compatible with chance. Official keyed leftover marked files still exceed 0.55 (**20/20** positive-control sensitivity; Table 9).

These results demonstrate paired-reference indication at the group grain within the studied setting for one public SynthID-Text instance. They do not establish calibrated standalone indication on an unfamiliar file (the method has not been validated or calibrated for unfamiliar web documents), a universal key-free detector, key recovery, or a refutation of Christ et al. (2024) or Zhang et al. (2024). A preregistered lengthening of the public mixin to $H_{\mathrm{w}} = 12$ (freeze SHA `b70986d`) is a cross-run contrast on the same prompt strings with new twins: interpolate last-4 ranking is **76/100** against lock A **99/100** on 100 one-liners, while official matching scores remain above 0.55 on all 400 marked files. Exact next-token occupancy is lower than under public $H_{\mathrm{w}} = 4$ (Table 13). The original-12 readout is **6/12** (imprecisely estimated at $n = 12$; clustered permutation $p = 0.247$). Isolated $\tau = 0$ on those 12 is 52/96. The original-12 hard matrix remains **25/48** sensitivity and **47/96** correct. On a Kirchenbauer green-list mixin with one-token seeding, interpolate last-4 ranks **12/12** (isolated **85/96**) on the original 12 strings and **100/100** (isolated 747/800) on 100 one-liners; that is a different construction, not a replacement of **25/48**.

Code and frozen JSON: `https://github.com/jensabrahamsson/text-watermark-laboratory` (pinned revision in Appendix A).

A Artifact identifiers and commands

Full dump paths and SHA-256 prefixes (first 16 hex digits) are listed in `paper/artifacts.json`. Identifiers below are short ids plus those prefixes; do not hyphenate them as English.

Software. Python $\geq$ 3.10; `transformers==4.57.6`; CPU JAX for official `detector_mean`; SynthID-Text public checkout installed with `pip install -e . --no-deps` in the public `synthid-text` checkout. Hugging Face ids: `gpt2`, `distilgpt2`, `Qwen/Qwen2-1.5B-Instruct`. Hub commit SHAs were not recorded at generation time. Published scores use the committed strings (Section 4.1); they do not re-download Hub weights. Current laboratory `gpt2` cache: `607a30d783dfa663caf39e06633` GitHub tree for this manuscript: `https://github.com/jensabrahamsson/text-watermark-laboratory/tree/1582a09` (100-family Kirchenbauer dumps). Start named `8f09aa6`.

Chronology (confirmatory 100). Protocol freeze SHA `7001489`. Prompts `experiments/2026-09-01-prompts-100/` committed in `294dba5` before `pair`. Pair SHA `bf98c92`. Analysis-code SHA `bbc802e` (truncated-context recount). Flags were not changed after the first run.

Headline commands. Variables: `PAIR12=experiments/2026-08-17-pair-12x4`, `PAIR36=experiments/2026-08-17-pair-36`, `PAIRDIR` is a new pair out-dir.

```
python -m
text_watermark_tools probe PAIR12 --methods hard --context-len 4
python -m text_watermark_tools probe PAIR36 --methods hits
python -m text_watermark_tools pair experiments/2026-09-01-prompts-100
  --n-samples 4 --max-new-tokens 128 --seed 20260901
python -m text_watermark_tools probe experiments/2026-09-01-pair-100x4
  --methods interpolate --context-len 4
python -m text_watermark_tools atoms --leave-one-out
  --test-dir experiments/2026-09-03-pair-12x4-ngram13
python -m text_watermark_tools pair PAIRDIR --hub-revision 607a30d783dfa663caf39e06633721c8d4cfcd7e
python -m text_watermark_tools pair experiments/2026-08-17-grok-prompts
  --n-samples 4 --max-new-tokens 128 --seed 20260904 --mixin kgw
  --hub-revision 607a30d783dfa663caf39e06633721c8d4cfcd7e
  --out-dir experiments/2026-09-03-pair-12x4-kgw
```

Distil/Qwen Phase B replace the model flag and run locks B and C only (`--methods poshits --fit-prefix 4 --pos-bucket 1`; `--rankpath --fit-prefix 4 --pos-bucket 1`).

Frozen result files. SHA-256 prefixes (first 16 hex digits) identify the JSON dumps. Full paths are in `paper/artifacts.json`. Original-12 hard: `experiments/2026-09-01-probe-12x4-recount-hard-last4/hard/holdout.json` (e4f758305b631761; 9/12, 25/48 vs 22/48, AUC 0.590). Lock A interpolate: `experiments/2026-09-01-probe-100x4-hard-last4/interpolate/holdout.json` (de0f0c5738729e87). GPT-2 lock B poshits: `experiments/2026-09-01-probe-100x4-opening-poshits/poshits/holdout.json` (aca0f852266c6b41). Distil lock B: `experiments/2026-09-01-probe-distil-100x4-opening-poshits/poshits/holdout.json` (37834c2f0e45856c). Qwen lock B: `experiments/2026-09-01-probe-qwen-100x4-opening-poshits/poshits/holdout.json` (5b7e3c00264561f8). Stem honesty: `experiments/2026-09-01-ranking-isolated-honesty/`. Absolute-history 12-LOO mask-m: `experiments/2026-09-03-probe-12x4-headline-windows-absolute/results.json` (58094d769726dc18; prefixes equal reindexed; hard tails 9/12). Longer-context protocol freeze: `b70986d` (`research/PROTOCOL-next-longctx.md`). Original-12 $H_{\mathrm{w}} = 12$ pair: `experiments/2026-09-03-pair-12x4-ngram13/results.json` (61153a82dab1eaea). Original-12 $H_{\mathrm{w}} = 12$ probe: `experiments/2026-09-03-probe-12x4-ngram13-hard-last4/hard/holdout.json` (fcff60687ac27348; interpolate and hard 6/12). Longer-history 100-family start (logbook, before generation): `facc538`. Longer-history 100-family pair: `experiments/2026-09-03-pair-100x4-ngram13/results.json` (761b5d2ea676fa5d). Longer-history 100-family probe: `experiments/2026-09-03-probe-100x4-ngram13-hard-last4/interpolate/holdout.json` (e17cda239aa36e43; interpolate 76/100). Occupancy command freeze: `df5487d`. Original-12 $H_{\mathrm{w}} = 12$ occupancy: `experiments/2026-09-03-atoms-12x4-ngram13/atoms.json` (4d29c92147e6da9d; 160 seen versus public 269). Original-12 public occupancy: `experiments/2026-09-03-atoms-12x4-public-loo/atoms.json` (1247287a6369e93d). 100-family $H_{\mathrm{w}} = 12$ occupancy: `experiments/2026-09-03-atoms-100x4-ngram13/atoms.json` (ee0fcb86e6aceafc; 5878 seen versus public 10158). 100-family public occupancy: `experiments/2026-09-03-atoms-100x4-public-loo/atoms.json` (6d6d516ec296aae1). Kirchenbauer mixin freeze: `research/PROTOCOL-next-kgw.md` (8371406). Original-12 pair: `experiments/2026-09-03-pair-12x4-kgw/results.json` (c4360e0c259b1f77). Original-12 probe: `experiments/2026-09-03-probe-12x4-kgw-hard-last4/interpolate/holdout.json` (67b8ed6bb2b96bcd; interpolate 12/12, 85/96). Original-12 occupancy: `experiments/2026-09-03-atoms-12x4-kgw/atoms.json` (cbde7405e481bfad; 114 seen). 100-family Kirchenbauer start: `8f09aa6`. 100-family pair: `experiments/2026-09-03-pair-100x4-kgw/results.json` (d96793694eaa2ab1). 100-family probe: `experiments/2026-09-03-probe-100x4-kgw-hard-last4/interpolate/holdout.json` (dce85a6796def442; interpolate 100/100, 747/800). 100-family occupancy: `experiments/2026-09-03-atoms-100x4-kgw/atoms.json` (e59753393fdc1d3e; 4557 seen). Claude 2026-09-03 resample report: `experiments/2026-09-03-resample-work/report.json` (e7a1a62a10585d1d; 40 texts). Premark vs new last-4: `experiments/2026-09-03-resample-work/blind-premark-vs-new-k4/results.json` (31894db2db724064; 35/40, `used_keys=false`). Previous vs new last-4: `experiments/2026-09-03-resample-work/blind-previous-vs-new-k4/results.json` (f852be4c7fd1dcf1; 29/40). Claude 2026-09-04 resample report: `experiments/2026-09-04-resample-work/report.json` (c54c5beb6da07adb; 40 texts). Premark vs new last-4: `experiments/2026-09-04-resample-work/blind-premark-vs-new-k4/results.json` (4a8f0fba8cbda791; 37/40, `used_keys=false`). Previous vs new last-4: `experiments/2026-09-04-resample-work/blind-previous-vs-new-k4/results.json` (b31a564d60ade346; 19/40). Distil Kirchenbauer freeze: `1540d3c`. Distil pair: `experiments/2026-09-03-pair-distil-12x4-kgw/results.json` (f17ba689c14ecf21). Distil interpolate: `experiments/2026-09-03-probe-distil-12x4-kgw-hard-last4/`

interpolate / holdout . json (845af54db1aeb37f; 12/12, 85/96). Distil occupancy: experiments / 2026-09-03-atoms-distil-12x4-kgw / atoms . json (52c8fe2505a567c0; 130 seen). Qwen Kirchenbauer freeze: 45a75c9. Qwen pair: experiments / 2026-09-03-pair-qwen-12x4-kgw / results . json (1e7e7a85888e2746). Qwen interpolate: experiments / 2026-09-03-probe-qwen-12x4-kgw-hard-last4 / interpolate / holdout . json (68b96241a91e8a9c; 12/12, 68/96). Qwen occupancy: experiments / 2026-09-03-atoms-qwen-12x4-kgw / atoms . json (df93d69c13d85869; 84 seen). Aaronson freeze: 747f3cd. Aaronson pair: experiments/ 2026-09-04-pair-12x4-aaronson / results . json (f1a8616c9b5ec347). Aaronson interpolate: experiments / 2026-09-04-probe-12x4-aaronson-hard-last4 / interpolate / holdout . json (9077e6d67f7c1900; 11/12, 56/96). Aaronson occupancy: experiments / 2026-09-04-atoms-12x4-aaronson / atoms . json (c1f16afa2ba1c418; 573 seen). 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Distil $H_{\mathrm{w}} = 12$ interpolate: experiments/2026-09-04-probe-distil-12x4-ngram13-hard-last4/ interpolate/holdout.json (e8ac790aebdb8919; 9/12, 49/96). Distil $H_{\mathrm{w}} = 12$ occupancy: experiments/2026-09-04-atoms-distil-12x4-ngram13/atoms.json (e0ccc7de1f47a79c; 175 seen versus 11994 unseen). Distil $H_{\mathrm{w}} = 12$ 100-family freeze: d891622. Distil $H_{\mathrm{w}} = 12$ 100 pair: experiments / 2026-09-04-pair-distil-100x4-ngram13 / results . json (b46a5d5debed1485). Distil $H_{\mathrm{w}} = 12$ 100 interpolate: experiments / 2026-09-04-probe-distil-100x4-ngram13-hard-last4 / interpolate / holdout . json (22d8c29006252a26; 88/100, 557/800). Distil $H_{\mathrm{w}} = 12$ 100 occupancy: experiments/ 2026-09-04-atoms-distil-100x4-ngram13/atoms.json (331dbab436097872; 11182 seen versus 85493 unseen). Qwen2-1.5B $H_{\mathrm{w}} = 12$ freeze: d7303a2. Qwen $H_{\mathrm{w}} = 12$ pair: experiments/ 2026-09-04-pair-qwen-12x4-ngram13 / results . json (905c76810744421d). Qwen $H_{\mathrm{w}} = 12$ interpolate: experiments/2026-09-04-probe-qwen-12x4-ngram13-hard-last4/ interpolate/holdout.json (617663de48b81879; 4/12, 41/96). Qwen $H_{\mathrm{w}} = 12$ occupancy: experiments/2026-09-04-atoms-qwen-12x4-ngram13/atoms.json (d7867f4c81b21ca2; 65 seen versus 12127 unseen). Distil Aaronson freeze: 9bdf12a. Distil Aaronson pair: experiments/ 2026-09-04-pair-distil-12x4-aaronson/results.json (ab8f1a9f340960c5). Distil Aaronson interpolate: experiments / 2026-09-04-probe-distil-12x4-aaronson-hard-last4 / interpolate / holdout . json (c36caf9745da2ce3; 7/12, 0/48 marked TP). Distil Aaronson occupancy: experiments / 2026-09-04-atoms-distil-12x4-aaronson / atoms . json (cc5ad2fcf035fdca; 196 seen versus 11996 unseen). Distil Aaronson 100-family freeze: bf05759. Distil Aaronson 100 pair: experiments / 2026-09-04-pair-distil-100x4-aaronson / results . json (a043578f9795a7be). Distil Aaronson 100 interpolate: experiments / 2026-09-04-probe-distil-100x4-aaronson-hard-last4 / interpolate / holdout . json (89d3b8c7f8d5d0e0; 96/100, 601/800). Distil Aaronson 100 occupancy: experiments / 2026-09-04-atoms-distil-100x4-aaronson/atoms.json (06b4d85a772626d4; 28824 seen versus 61305 unseen). Qwen Aaronson freeze: 1171d5c. Qwen Aaronson pair: experiments/ 2026-09-04-pair-qwen-12x4-aaronson/results.json (12ea3ef1c34f037b). Qwen Aaronson interpolate: `experiments / 2026-09-04-probe-qwen-12x4-aaronson-hard-last4 / interpolate / holdout . json` (`ac41821f88adba14`; 12/12, 12/48 marked TP). Qwen Aaronson occupancy: `experiments / 2026-09-04-atoms-qwen-12x4-aaronson / atoms . json` (`419a2088b2ba8e6e`; 457 seen versus 11735 unseen). Qwen2-1.5B $H_w = 12$ 100-family freeze: `636765c`. Qwen $H_w = 12$ 100 pair: `experiments / 2026-09-04-pair-qwen-100x4-ngram13/results.json` (`ca4c793eaaf77c18`). Qwen $H_w = 12$ 100 interpolate: `experiments / 2026-09-04-probe-qwen-100x4-ngram13-hard-last4 / interpolate / holdout . json` (`d051137c566c5629`; 76/100, 474/800). Qwen $H_w = 12$ 100 occupancy: `experiments / 2026-09-04-atoms-qwen-100x4-ngram13 / atoms . json` (`1d0ae9837b3cd4e0`; 3535 seen versus 98064 unseen). Qwen Aaronson 100-family freeze: `a761a7d`. Qwen Aaronson 100 pair: `experiments/2026-09-04-pair-qwen-100x4-aaronson/ results . json` (`4fb67051fb89839b`). Qwen Aaronson 100 interpolate: `experiments / 2026-09-04-probe-qwen-100x4-aaronson-hard-last4 / interpolate / holdout . json` (`9bb1cf87dc11328e`; 100/100, 616/800). Qwen Aaronson 100 occupancy: `experiments / 2026-09-04-atoms-qwen-100x4-aaronson / atoms . json` (`d3932e3b1346789b`; 8750 seen versus 92842 unseen). Qwen Kirchenbauer 100-family freeze: `ed9fb20`. Pair out-dir: `experiments / 2026-09-04-pair-qwen-100x4-kgw`. Official first-draw $z > 3$ **90/100**; interpolate **96/100** (isolated **620/800**). That lock is not **25/48**. Distil / gpt2-medium unmarked-LM opening rankpath freeze: `d8e6f7f`. Distil-LM isolated 32/48 vs 31/48 (ranking 10/12). gpt2-medium-LM isolated 31/48 vs 32/48. Those isolated counts are not **25/48**. Twins: `experiments / 2026-08-17-pair-12x4`. Out-dirs: `experiments / 2026-09-04-probe-12x4-rankpath-distil-lm`, `experiments / 2026-09-04-probe-12x4-rankpath-medium-lm`.

Internal labels, on first use in the repository. *Grok-length*: original 12 scene prompts, comparable length to a short narrative seed. *Stem*: prompt-family id. *Opening-atom overlap*: an evaluation next-token that already occurred under the same short context in training.

B Additional transfer catalog

Transfer and leftover catalog. These rows are not headlines (isolated-headline scope).

Out-of-family lock A on the original 12. Tables from the 100 one-liners, nested Youden (post hoc) 23/48 vs. 38/48 (prompt 8/12). Occupancy-free postokhits at $\tau = 0$ is 16/48 vs. 48/48. Opening coverage is 18/48. Low-recall, high-specificity readers can have higher balanced accuracy than 47/96 without being a better prespecified detector: thresholds, dependence, and abstentions differ. They are not substituted for the hard $\tau = 0$ matrix.

Same-register 36. 100→36 lock A nested Youden 109/144 vs. 122/144 (prompt 36/36). Same register, not isolated detection on the original 12.

Register mismatch. Grok-length train → original 12 lock A nested 16/48 vs. 41/48. Reverse 100→Grok-register 12 lock A nested 22/48 vs. 41/48; occupancy-free 0/48. The original 12 are not a unique failure set.

Mask-m. Masking the first scored tokens of 12-LOO hard last-4 does not destroy prompt ranking. Table 14 is the absolute-history remasure (protocol SHA `b70986d`; reindexed dumps stay). Prefix [0:4) and hard tails stay at the reindexed ranks; interpolate [8:128) rises 3 → 4. Hard [4:128) leftover is 11/20 vs. 11/20. Isolated $\tau = 0$ stays compatible with chance on every slice; the original-12 matrix remains **47/96**.

Table 14: Absolute-history 12-LOO mask-m last-4, original 12×4. Absolute dump SHA prefix `58094d769726dc18`. Reindexed dump stays. Isolated $\tau = 0$ is not **25/48**.

Window	Hard abs.	Hard reind.	Interp. abs.	Interp. reind.
[0:2)	10/12	10/12	9/12	9/12
[0:4)	5/12	5/12	9/12	9/12
[0:8)	6/12	6/12	8/12	8/12
[2:128)	9/12	9/12	5/12	5/12
[4:128)	9/12	9/12	5/12	5/12
[8:128)	9/12	9/12	4/12	3/12

Longer hash history. Main-text Section 9: $H_\mathrm{w} = 12$ interpolate ranking is **76/100** on 100 one-liners and **6/12** on the original 12 strings. Official matching scores remain above 0.55. Group ranking **76/100** is below lock A **99/100**. The public $H_\mathrm{w} = 4$ headline remains **47/96**; the new $H_\mathrm{w} = 12$ twins yield isolated 52/96. Exact next-token occupancy is lower under $H_\mathrm{w} = 12$ than under public $H_\mathrm{w} = 4$ (Table 13).

Control keys. Text sampled with `control-shuffled-30` scores near 0.50 on the public keys (mean of 48 file means 0.501) and near 0.62 on the matching control keys (0.624). Key-free opening poshits trained on public twins assign $\Lambda > 0$ to 0/48 control files. Public vs. control still ranks 12/12 (AUC 0.906). The reader is instance-specific without recovering keys.

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